Factor-base designs: A11 replication on p_2 = 2013265921 (run-20261015T2307Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. **Protocol:** PREREGISTRATION.md §5, with Amendment 2 F1 (registered on `research/factor-base-designs-lean` in commit `e0afa082d`, before any p_2 row) and Amendment 1 for everything F1 does not change. This is F1 only. F2–F4 were not run here.

Run

- **Command:** `target/release/examples/fb_designs_slope run --p 2013265921 --arms A0,A11 --m 2,3 --out-dir research/factor_base_designs_20261015/runs/p2` (defaults: t from 2, at most 27 = $v_2(p_2 - 1)$, 2 planted targets per cell, F4 budget 300 s, hard kill 420 s, `ulimit -v 13000000` per child).
- **Commit measured:** `d5f675260867f2d025e080e65d8d7027431bc845`, recorded in every row.
- **Host:** "vm", Intel Xeon @ 2.10 GHz, 4 cores, a Claude Code cloud container. This is a different container and CPU from the p_1 run (2.80 GHz), so times are not comparable across the two runs. **Isolation:** one system at a time, no other work running.
- **Window:** 2026-10-15 23:07Z to 2026-10-16 about 00:08Z (UTC).
- **Seeds (Amendment 2):** `0x46424453 ⊕ prime≪48 ⊕ arm≪40 ⊕ m≪32 ⊕ t≪16` with p_2 = 1. Target k is planted from `cell ⊕ (k+1)≪8`. With p_1 = 0, the p_1 seeds are unchanged: A0 and A11 at m = 3, t = 3 were rebuilt and match their committed rows (x_R, D, max_cols).
- **Records (after reconciliation, see below):** every system this run measured is in the F1 rows of record `runs/F1-{A0,A11}-m{2,3}-p2.jsonl`, except A11 m = 2, t = 10 and t = 11, which are in `runs/F1-annex-p2/A11-m2-p2-t10-t11.jsonl`. The annex also holds this run's logs and the row-by-row cross-check (`crosscheck.txt`). The rows as first committed are at `runs/p2/` in commit `cb750c828`, with `analysis.txt` and `instrument-check-p2.txt`.

F1 pre-check

$p_2 < 2^{31}$, and `f4_fp` reduces modulo $p < 2^{32}$. Eight planted systems at t = 2 were built before any p_2 row: A0 and A11, m = 2 and 3, both targets. Each passed the set, presentation and vanishing checks, and each kept its planted solution in F4's basis (`planted_ok` true, not inconsistent). The check was recorded in `runs/p2/instrument-check-p2.txt` (commit `cb750c828`). The parent lane's own F1 pre-check is `runs/f1-instrument-check-p2.txt`.

Graphs

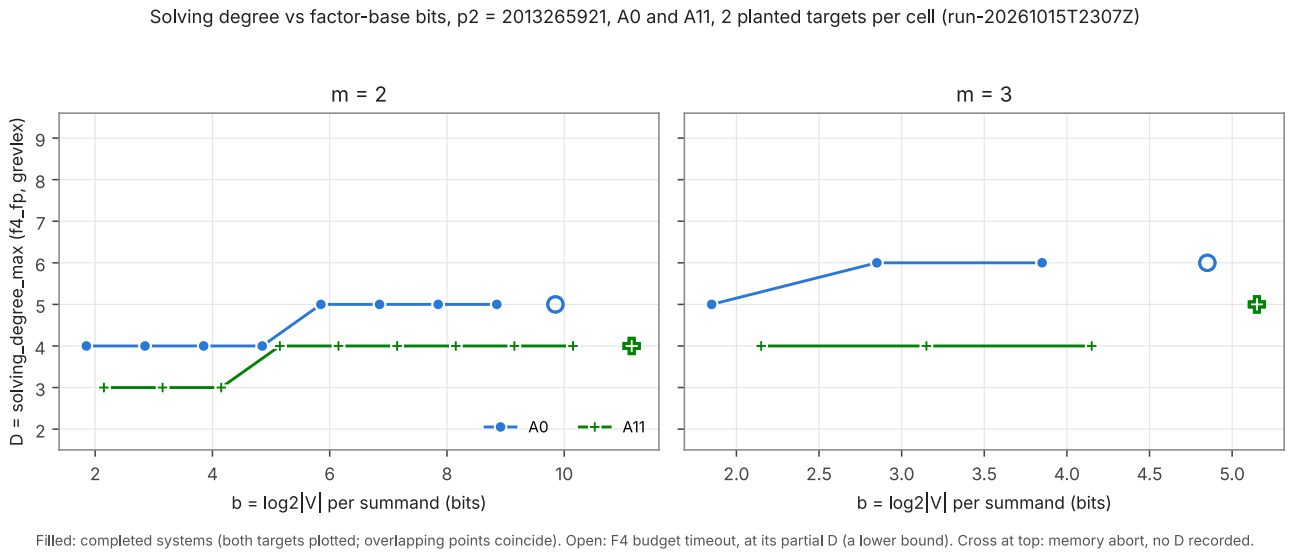


Figure 1: Measured D against b on p_2, A0 and A11, m = 2 (left) and m = 3 (right). Both targets of each cell are plotted (n = 2 per cell; they coincide). Open markers are systems stopped by the 300 s F4 budget, at their partial D. Rows from `runs/F1-*-p2.jsonl` and the annex, per `plot.py`.

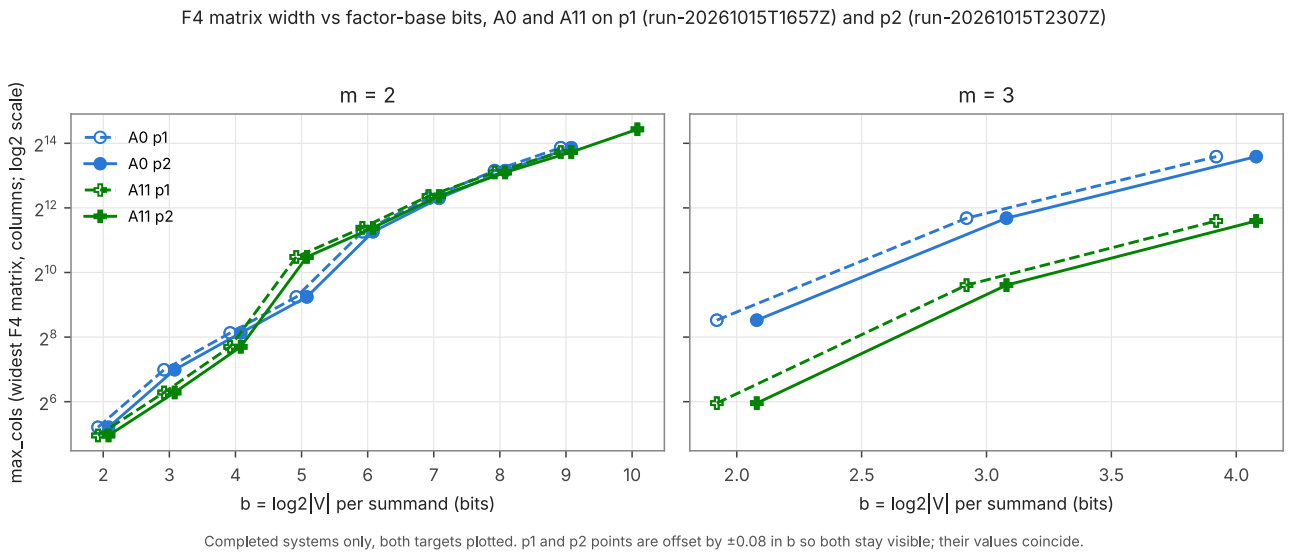


Figure 2: Widest F4 matrix (`max_cols`, $\log_2$ scale) against b, A0 and A11 on p_1 (dashed, hollow) and p_2 (solid). The points are offset ±0.08 in b so both primes stay visible. Completed systems only.

F1 decision rule (registered for m = 3 on p_2)

"A11 replicates if A11's cell D ≤ A0's cell D − 1 and A11's `max_cols` < A0's at every shared completed t, with at least 3 shared completed cells." The cell `max_cols` here is the mean over the cell's two systems.

t	A11 D	A0 D	A11 max_cols	A0 max_cols	condition
2	4	5	62	368	met
3	4	6	781	3285	met
4	4	6	3096	12314	met

Reading: replicates, on 3 shared completed cells, the minimum the rule accepts. At t = 5 both arms' systems hit the F4 budget, so that cell is missing for both, with partial D of 5 (A11) and 6 (A0) when stopped.

σ on p_2 (from `analysis.txt`): A0 m = 2 is 0.1905 (8 cells), A0 m = 3 is 0.5000 (3), A11 m = 2 is 0.1500 (9), and A11 m = 3 is 0.0000 (3). The registered width-growth slope $\Delta\log_2$ `max_cols`/Δb (descriptive) is 1.261, 2.532, 1.215 and 2.821, in the same order.

Observations

- **p_1 and p_2 agree system for system.** 44 planted systems completed on both primes at the same (arm, m, t, target): A0 and A11, m = 2 and 3. All 44 have the same D, and all 44 have the same `max_cols`. The two primes share no curve, set or target. At these sizes, D and matrix width do not vary with the prime or the instance.
- **m = 2 (descriptive; the rule is not registered here).** A11's D is one below A0's at t = 2–4 and 6–9, and equal at t = 5. A11 is narrower at t = 2–4 and 8–9, and wider at t = 5–7. The widest of those is t = 5: 1431 vs 606 columns.
- **Further cells.** A11 at m = 2 completed one more cell on p_2 (t = 10, D = 4, about 257 s per system) than on p_1, where t = 10 timed out at 300 s on a different CPU. A0 at m = 2 stops at t = 10 on both primes.
- **The two targets of a cell are not always distinct systems.** At small t only 2–4 elements of V lie on the curve, so both targets can draw the same abscissae. Then x_R coincides, and for the tower arms the system is identical. On p_2 this happened in A11 at t = 2 (m = 2 and 3). On p_1, eight cells at t ≤ 4 across arms have it. D is unaffected, but those cells hold one distinct system, not two. The draw rule is the registered one and was not changed.

Every system (p_2)

Target 0 / target 1. Time is F4 wall seconds. Status is ok or timeout (F4 budget of 300 s). There were no memory aborts and no planted-solution violations.

arm	m	t	vars	eqs	D	max_cols	time (s)	status
A0	2	2	4	5	4 / 4	37 / 37	0.0 / 0.0	ok, ok
A0	2	3	6	7	4 / 4	127 / 127	0.0 / 0.0	ok, ok
A0	2	4	8	9	4 / 4	280 / 280	0.0 / 0.0	ok, ok
A0	2	5	10	11	4 / 4	606 / 606	0.0 / 0.0	ok, ok
A0	2	6	12	13	5 / 5	2451 / 2451	0.2 / 0.2	ok, ok
A0	2	7	14	15	5 / 5	5031 / 5031	1.1 / 1.2	ok, ok

A0	2	8	16	17	5 / 5	9119 / 9119	9.3 / 9.0	ok, ok
A0	2	9	18	19	5 / 5	14922 / 14922	59.6 / 59.4	ok, ok
A0	2	10	20	21	5 / 5	61361 / 61361	301.1 / 301.5	timeout, timeout
A0	3	2	7	8	5 / 5	368 / 368	0.0 / 0.0	ok, ok
A0	3	3	10	11	6 / 6	3285 / 3285	0.3 / 0.3	ok, ok
A0	3	4	13	14	6 / 6	12314 / 12314	9.5 / 9.7	ok, ok
A0	3	5	16	17	6 / 6	97456 / 97456	301.0 / 301.3	timeout, timeout
A11	2	2	9	10	3 / 3	31 / 31	0.0 / 0.0	ok, ok
A11	2	3	12	13	3 / 3	78 / 78	0.0 / 0.0	ok, ok
A11	2	4	15	16	3 / 3	208 / 208	0.0 / 0.0	ok, ok
A11	2	5	18	19	4 / 4	1431 / 1431	0.0 / 0.1	ok, ok
A11	2	6	21	22	4 / 4	2673 / 2673	0.5 / 0.6	ok, ok
A11	2	7	24	25	4 / 4	5282 / 5282	2.4 / 2.3	ok, ok
A11	2	8	27	28	4 / 4	8763 / 8763	10.6 / 10.4	ok, ok
A11	2	9	30	31	4 / 4	13579 / 13579	53.9 / 53.7	ok, ok
A11	2	10	33	34	4 / 4	22118 / 22118	259.5 / 255.7	ok, ok
A11	2	11	36	37	4 / 4	184920 / 184920	302.0 / 301.9	timeout, timeout
A11	3	2	15	16	4 / 4	62 / 62	0.0 / 0.0	ok, ok
A11	3	3	20	21	4 / 4	781 / 781	0.0 / 0.1	ok, ok
A11	3	4	25	26	4 / 4	3096 / 3096	1.3 / 1.3	ok, ok
A11	3	5	30	31	5 / 5	40486 / 40486	300.8 / 301.6	timeout, timeout

Requested and completed

4 series, 27 cells, 54 planted systems: 46 completed and 8 stopped at the F4 budget (A0 m = 2 t = 10, A0 m = 3 t = 5, A11 m = 2 t = 11, A11 m = 3 t = 5, both targets each). The commit that landed these rows (`c195dfa64`) states “50 systems, 42 completed” in its message. That was a miscount; these figures are correct.

Reconciliation with the parent lane (2026-10-16)

The parent lane ran F1 on its own, earlier (commits `f287673fe` and `fa60aba73`), with the same seed rule. `fb_designs_slope compare-rows` compared the two runs system by system (`runs/F1-annex-p2/crosscheck.txt`). Of 52 shared systems, 50 are identical in construction and F4 output. The other 2, A11 m = 2 t = 10, have identical construction, but the parent’s run timed out at 300 s and this run completed (D = 4 in both). This run also has A11 m = 2 t = 11 (timeouts).

The parent’s files are the F1 rows of record. This run’s duplicate rows were removed, and its extra rows were kept in the annex. Every number in this report is unchanged, and the graphs rebuild byte-identical from the new sources. The F1 verdict (m = 3: replicates) rests on m = 3 rows that are identical in both runs. The σ values above use this run’s rows. With the rows of record, A11 m = 2 on p_2 reads σ = 0.1786 on 8 cells, instead of 0.1500 on 9; both are “candidate”.

Deviations

None from Amendment 2 F1. Not run in this lane: F2 (the N11 null), F3 (the m = 3, t = 5 cell at 3600 s) and F4 (m = 4). The parent lane ran all three (`runs/analysis-amendment2.txt`). F2 reads “shape, not structure”: the A11-shaped null has A11′s D in every completed cell. So the D difference this run replicates comes from the shape of A11′s equations, not from the curve.